ENG-SCI 51 Report
Team 1: Squirrels

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Figure 1: Final Robot with Controller and 2nd Place Medal

1 Abstract

To learn about manufacturing, computer aided design, project integration, and robotics, the team manufactured and developed a robot to compete at the Turf Wars. The robot's goal is to be able to traverse 15° and 30° inclines, grab wooden blocks, tennis balls, and dog toys, and lift them to shelves at 7, 14, and 21 inches in height. To solve this problem, a rotating arm with a claw attached at the end was built, inspired by a squirrel's tail; pulley systems, gears, and motors were used. The design is a practical solution whose most challenging aspect is ensuring that the arm does not snap under tension. The main strategic objectives of reaching the highest shelf, and reducing as much as possible the time to acquire were both fulfilled. This project taught us how to interpret and produce technical drawings, to model in 3D using SolidWorks, to perform varied manufacturing operations, to link physics to reality, and to assemble a complex project.

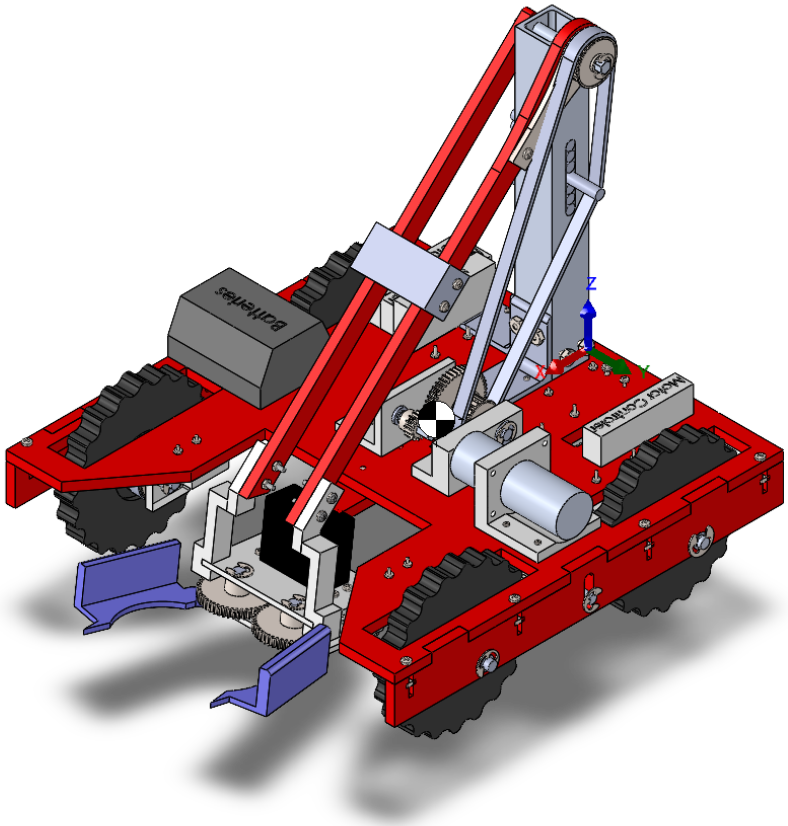


Figure 2: Final CAD Model in SolidWorks

2 Concept Development

2.1 Constraints

There were several constraints that limited the robot design:

Size

The robot is expected to fit within an 11”x11”x11” box. This meant that we had to design for a robot to fit within those dimentions initially, but then expand to be able to reach the 21 inch height.

Materials

The amount of material to be used was limited and it was prohibited to acquire any extra material. This had the goal of ensuring fairness for all teams, prevent the acquisition of costly or overly sophisticated components, and ensure an efficient use of the materials.

Time

The team had to design, manufacture, prototype, test, and complete the robot within a time period of 3 months beginning September 3rd 2025 and ending December 3rd 2025.

2.2 Success Criteria

Acquisition Speed

The robot should be able to acquire a 4” round dog toy within 10 seconds of the game starting.

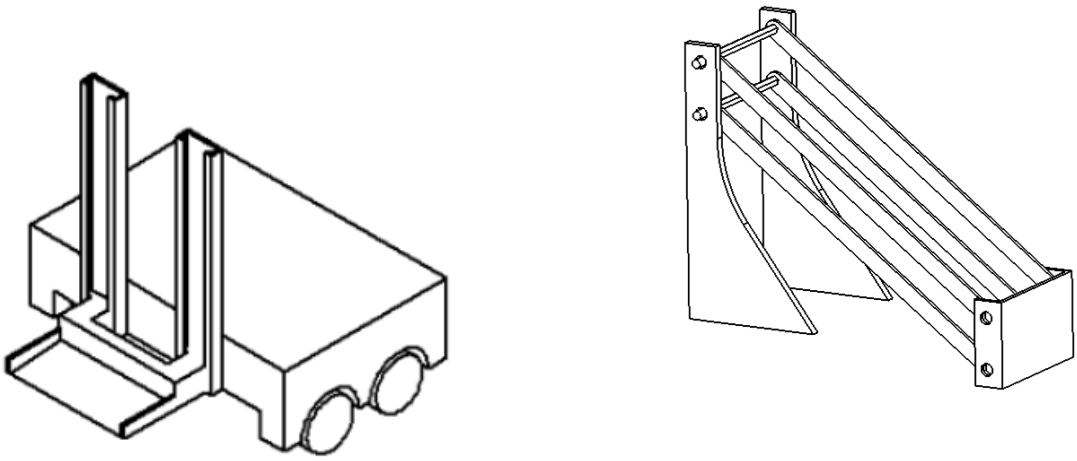
Lifting

The robot should be able to efficiently lift at least two types of objects: tennis balls and dog toys. It must also be able to lift the dog toys to a height of 21 inches without tipping. This would allow the robot to get a maximum of 17 points.

Movement

The robot should be able to traverse the 15° and 30° inclines without tipping or becoming high centered.

2.3 Alternative Solutions



(a) Forklift (b) Four Bar Linkage

Figure 3: Alternative Lifting Options

Pugh Matrix

Criteria	Design #1 (Forklift)	Design #2 (Four Bar Linkage)	Design #3 (The Squirrel's Tail)
Agility of Robot	+1	0	0
Precision	+1	+1	0
Gear power	0	0	-1
Tolerance	0	-1	+1
Time + Effort	0	-1	0
Time to Acquire	0	-1	+1
Ability to pick up multiple blocks	+1	-1	+1
Reach of Arm	-1	0	+1
Net Score	**+2**	**-3**	**+3**
Rank	**2**	**3**	**1**
Continue?	No	No	Yes!

Table 1: Comparison of Three Lifting Design Options

Analysis of Alternative Solutions

Three designs for the lifting mechanism were initially considered. The Forklift (fig. 3a), the Four Bar Linkage (fig. 3b), and the Squirrel's Tail (a two bar linkage which became our robot, fig. 2). Given the high point values assigned to the toys that required a greater height, the Squirrel's Tail was a clear winner because it could extend to a height of 21 inches, unlike the Forklift. In addition, it would be easier to manufacture than the Four Bar Linkage since it would require fewer components. Moreover, it was thought that the Squirrel's Tail could potentially lift multiple objects (this was false), which would have given it a significant edge.

3 Analysis

3.1 Drivetrain

The drivetrain (fig. 4) consisted of a four wheel drive design powered by 2 motors placed low and in the rear of the bot. This design required the use of belt to drive the forward wheels, but allowed for essentially zero turning radii and high maneuverability on tricky terrain. It was extremely pertinent that to solve for the appropriate torque needed by our motors to create the force required to enable the robot to surmount both ramps.

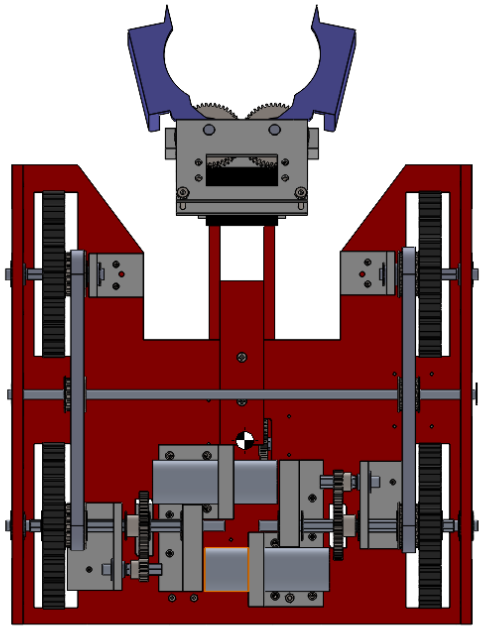


Figure 4: Bottom View of Bot Highlighting the Drivetrain

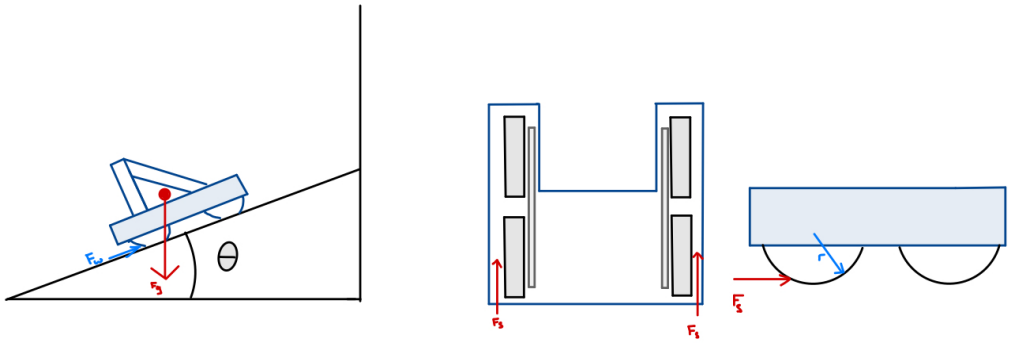


Figure 5: FBD for Tractive Force and Tourque Calculations

$$m = 2.57 \text{ kg}, \qquad d = 3.828 \text{ in} = 3.828 \times 0.0254 \approx 0.0972 \text{ m},$$
$$r = \frac{d}{2} \approx 0.0486 \text{ m}, \quad g = 9.81 \text{ m/s}^2$$

$$F_{\text{hill}} = mg \sin\theta$$

$$F_{\text{side}} = \frac{F_{\text{hill}}}{2} = \frac{mg \sin\theta}{2}$$

$$\tau_{\text{motor}} = F_{\text{side}} r = \frac{mgr}{2} \sin\theta$$

$$\theta = 15^\circ: \quad F_{\text{hill}} \approx 6.53 \text{ N}, \quad \tau_{\text{motor}} \approx 0.159 \text{ N} \cdot \text{m},$$
$$\theta = 30^\circ: \quad F_{\text{hill}} \approx 12.6 \text{ N}, \quad \tau_{\text{motor}} \approx 0.306 \text{ N} \cdot \text{m}.$$

3.2 Idealized Model

The idealized model calculations determine the maximum moment one side of the base plate can exert given that we pick up one dog toy and attempt to place it at the top shelf.

Given Values

$$\begin{aligned}
\text{Acrylic density:} &\quad 1.18\,\text{g/cm}^3 = 1180\,\text{kg/m}^3 \\
\text{Acrylic volume:} &\quad 6.70 \times 10^{-5}\,\text{m}^3 \\
\text{Acrylic force:} &\quad 1180 \times 1.10 \times 10^{-4} \times 9.81 = 1.272\,\text{N} \\
L &= 10.5\,\text{cm} = 0.105\,\text{m} \\
\text{Acrylic force per meter:} &\quad 4.7844\,\text{N/m}
\end{aligned}$$

Model Description

The beam is modeled as a rigid body of length $L = 0.105\,\text{m}$ with a uniform distributed load $w(x)$ representing the weight of the acrylic, and a single downward point load F_{dogtoy} representing the dog toy. The reaction moment at point A is M_A.

Functions

$$w(x) = 4.78\,\langle x \rangle^0 \qquad \text{(weight of acrylic)}$$
$$D(x) = -4.78x - F_{\text{dogtoy}}\langle x - 0.13335 \rangle^0 \qquad \text{(dog toy load)}$$
$$M(x) = -\frac{4.78x^2}{2} - F_{\text{dogtoy}}\langle x - 0.13335 \rangle' \qquad \text{(reaction moment)}$$

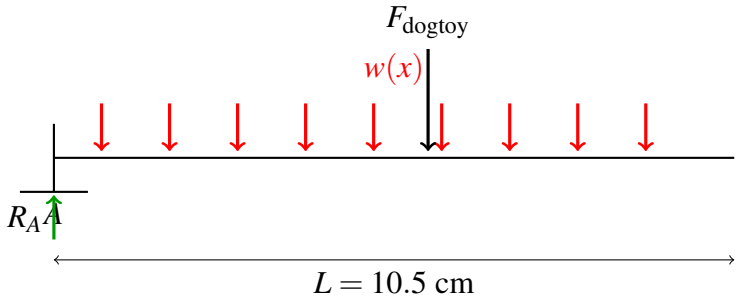


Figure 6: Idealized beam model with uniform acrylic weight and dog-toy point load.

Moment Balance

$$M_A = 2.39x^2 + 0.0567 \times 9.81 \times \left(x - \frac{10.5}{2}\right)$$

The maximum moment occurs when $x = L = 10.5\,\text{cm}$.

$$\begin{aligned} M_A &= 2.39x^2 + 0.0567 \times 9.81 \times \left(x - \frac{10.5}{2}\right) \\ &= 2.39(0.2647)^2 + 0.0567 \times 9.81 \times (0.2647 - 0.1055) \\ &= 0.24417 \end{aligned}$$

$$\text{therefore} \quad M_A \leq 0.24417\,\text{N·m}$$

Conclusion

This result means that the base plate will experience a maximum bending moment of approximately $0.244\,\text{N·m}$ when lifting the dog toy. Through calculating this, we establish how severely the base plate is loaded during the pick-up of the dog toy and can treat this value as a design benchmark for calculating moments with different competition items. Specifically, because this value of torque is much smaller than that of the arm and the drivetrain, it suggests that the base plate is not a critical point of concern. In addition, when factoring the area of the base plate into consideration, the stress of the base plate is also relatively low as we know stress = moment/section modulus, where section modulus is the moment of inertia divide by the maximum cross-section length of the base plate. As a result of the large base plate, the stress will be relatively low compared to the arm and the claw, and should have the bending capacity to all competition-related robot drivetrain activities.

3.3 Arm

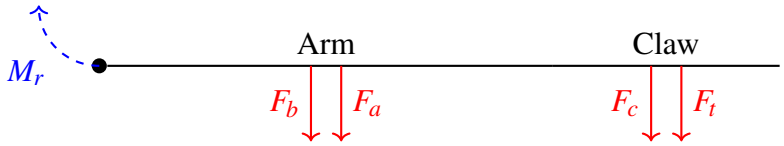
One of the most important tasks was to ensure that the arm could turn and lift objects. Consequently, the worst case analysis technique was used, evaluating how much torque the arm would require to support a tennis ball at 90° (maximizing torque). The resulting moment would be the sum of the torques.

$$F = m * 9.81 \tag{1}$$

$$T = F * d \tag{2}$$

$$M_r = \sum T \tag{3}$$

In this case, we had 4 torques: arm bracket, arm, claw, tennis ball.



$$M_r = \sum T = T_b + T_a + T_c + T_t \tag{4}$$

$$\sum T = F_b * d_b + F_a * d_a + F_c * d_c + F_t * d_t \tag{5}$$

$$\sum T = 9.81 * m_b * d_b + 9.81 * m_a * d_a + 9.81 * m_c * d_c + 9.81 * m_t * d_t \tag{6}$$

$$\sum T = 9.81 \frac{m}{s^2} * 0.00136Kg * 0.137m + 9.81 \frac{m}{s^2} * 0.054Kg * 0.136m + 9.81 \frac{m}{s^2} * 0.508Kg * 0.323 \tag{7}$$

$$+9.81 \frac{m}{s^2} * 0.0567Kg * 0.33m \tag{8}$$

$$M_r = 0.4188Nm \tag{9}$$

As a whole, this means that the arm would need 0.4188Nm in order to be able to withstand the torques. Consequently, our motors were characterized to determine whether they could produce enough torque.

As shown by the line 'Mr' in purple in Figure 7 in the Beam analysis section, the robot requires 1.6V. And even with that voltage the gear train and pulley system will need to have a high efficiency (close to 90 per cent).

3.4 Beam Analysis

We performed a Finite Element Analysis in SolidWorks to analyze the stress and displacement of our beam in addition to the pertinent shear force and bending moment diagrams. The maximum stress occurs at the lower right edge while the maximum displacement also occurs at that place. This corresponds to the point where the claw is attached to, and thus we reinforced the joint with proper bearings. Additionally, the top of the beam was re-designed to include a bigger circular section to distribute stress better.

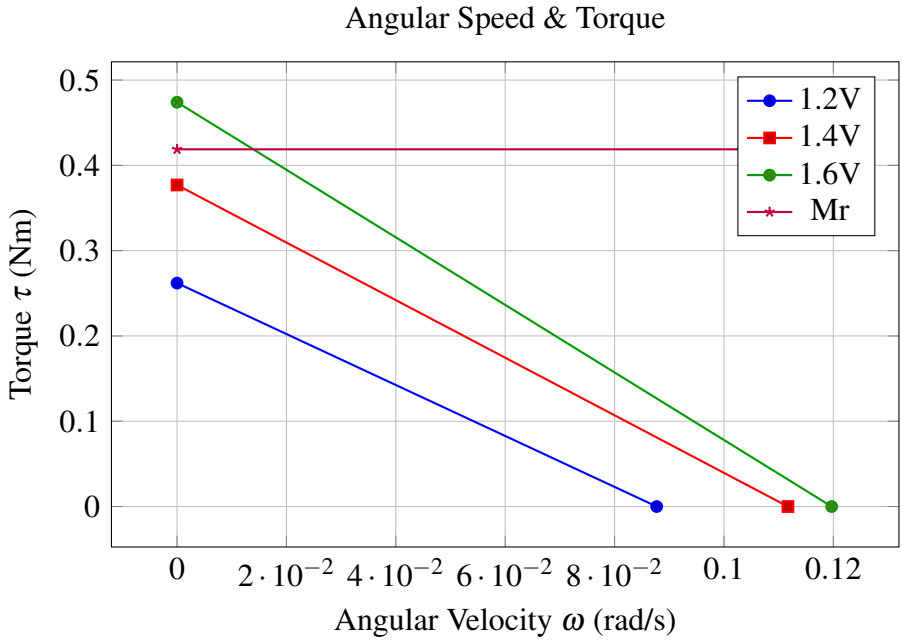


Figure 7: Motor performance at different input voltages

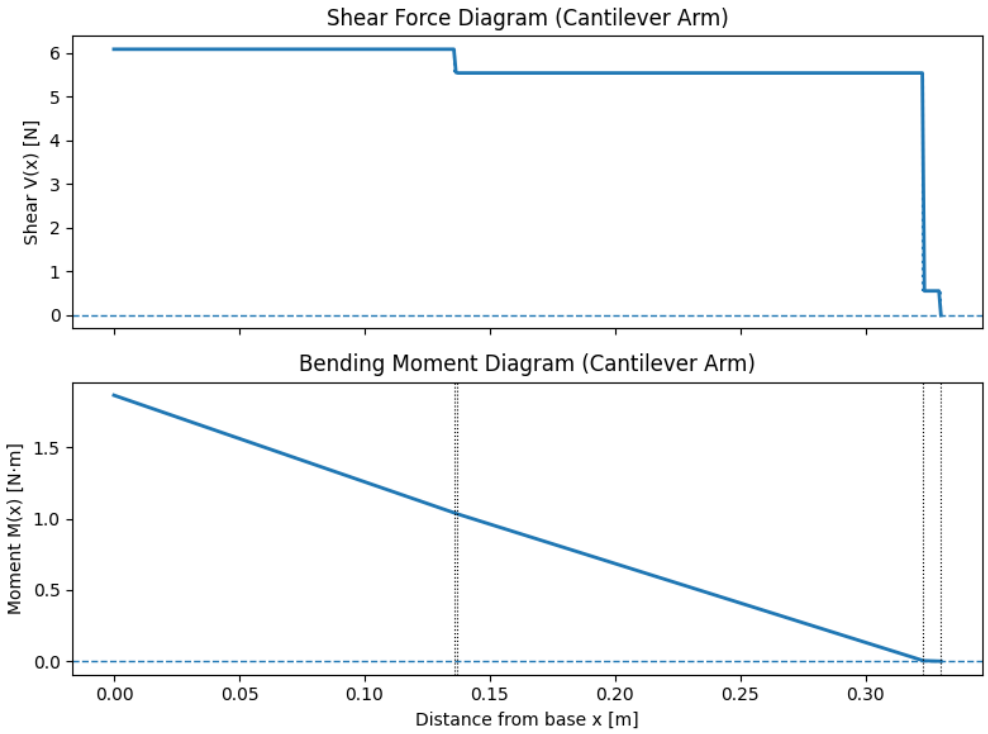


Figure 8: Shear Force and Bending Moment

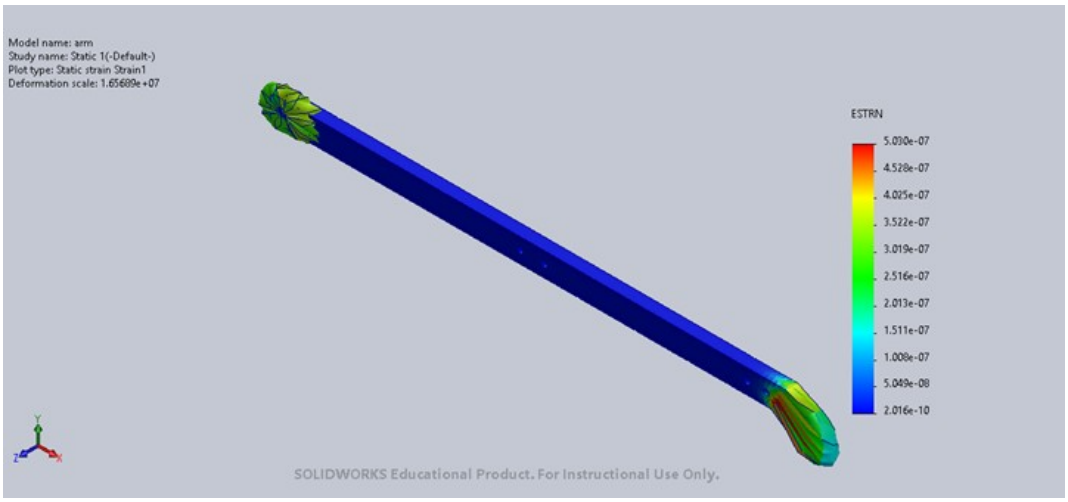


Figure 9: Finite Element Analysis

3.5 Claw

Our first claw design consisted simply of a scoop that could open or close. This depended entirely from the servo's strength, and our gear ratio was set. So, instead of trying to provide more strength to the claw, a redesign was needed. Knowing that the dog toys would be the robot's main objective, and that subsequently it could attempt to launch tennis balls, it was designed for optimizing the grip of those two objects.

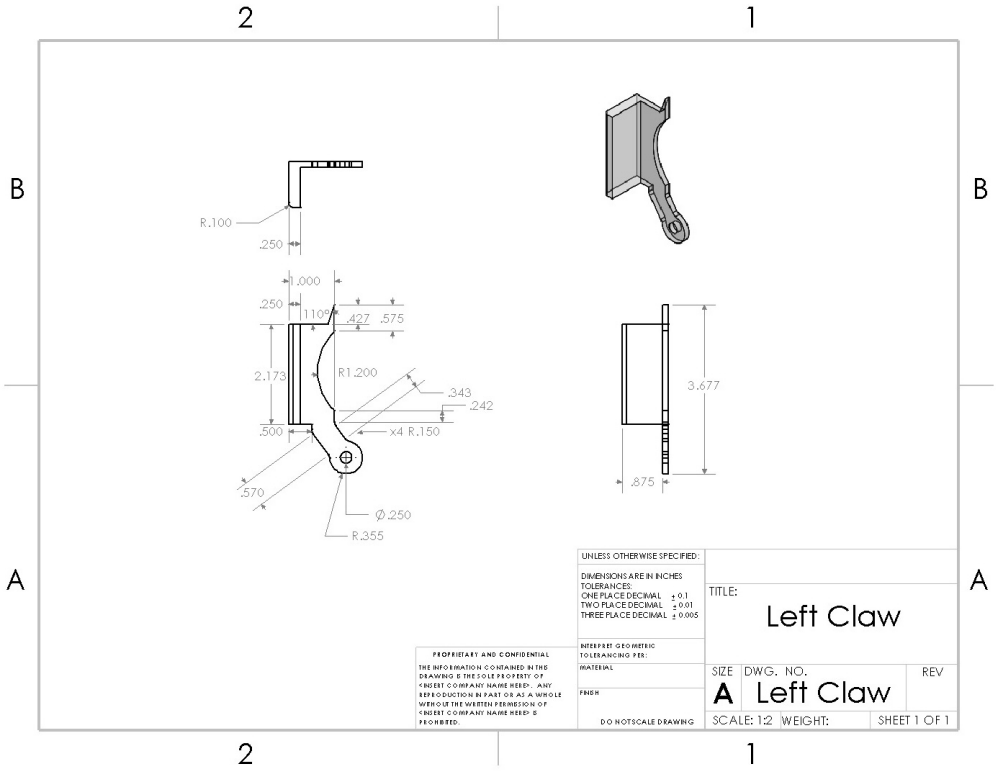


Figure 10: Revised Claw design

Consequently, to ensure that the robot would be able to pick up the dog toys pointy edges were incorporated to take advantage of the hollow nature of dog toys. This ensured that the robot would be able to lift them disregarding the servo's strength. Lastly, the claw design has a curved section which has the radius of the tennis ball to ensure a proper grip independently of the claw's strength.

3.6 Final Solutions

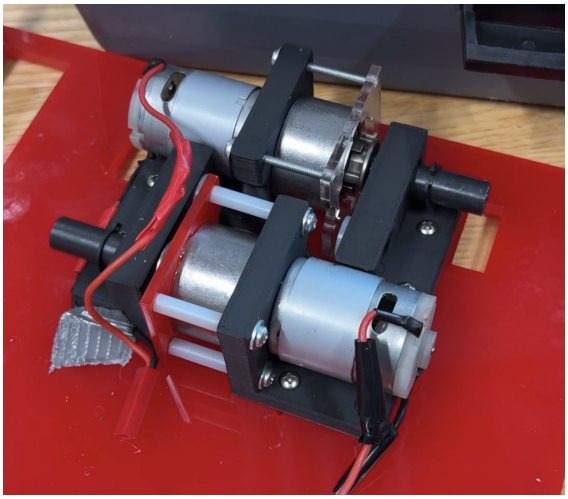
The Squirrel's Tail

The Squirrel's Tail acted exactly as we intended, allowing the robot to reach a height of 21 inches holding a dog toy. Ultimately, the Squirrel's Tail allowed for a maximum in precision and vertical reach.

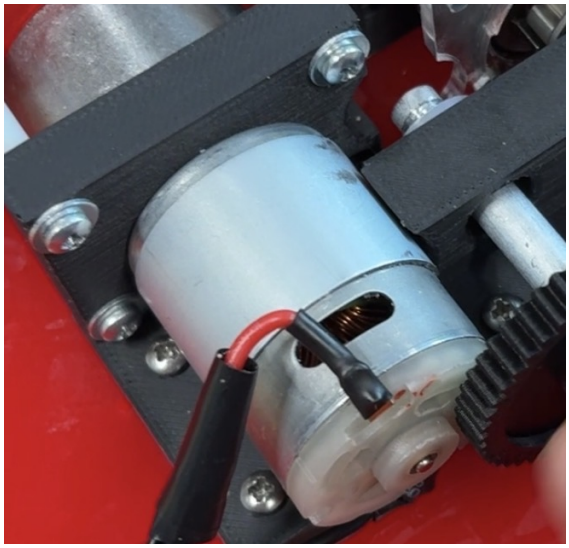
Claw

The claw (fig. 10) was fully functional, but could have used a larger gear ratio to improve its grip strength. The strength of the provided servo was severely overestimated, and thus the claw dropped a few objects during play. This issue was partially solved by adding a curve to the claw, which allowed it to better grasp tennis balls, but still continued to cause problems.

Compact Drivetrain



(a) Compact Drivetrain



(b) Detail: High Tolerance

Figure 11: Compact Drivetrain and Detail

We designed a compact drivetrain (fig. 11a) by making custom motor/shaft supports (fig. 11b), which successfully kept the center of gravity directly under the mast. This was crucial when the robot reached up-and-over to deposit dog toys and tennis balls, as well as drove down ramps. With a further back center of gravity, the robot may have tipped.

Overall Reflection

In general, the robot met the success criteria we established. It fit in the Box of Justice, could acquire an object in under 10 seconds, lifted dog toys to a height of 21 inches without tipping, and easily climbed both the 15° and 30° ramps. It was the lightest, highest-reaching, and fastest solution we developed, with the fewest limitations. (Technically, it could not pick up the blocks, but they were not worth many points.)

In the competition, it performed very well overall, but the lifting mechanism failed in the final round. Band stretch proved to be a real-world issue; although we believed we had accounted for it, our tensioning device was not as easily serviceable as we expected. In hindsight, we should have taken the time to machine another threaded tensioning rod, as we did for the drivetrain. We also had several instances where the claw failed to apply enough inward torque, suggesting that additional gearing in the claw's mechanism would have improved reliability.

If we were to approach the problem again, we would design a larger gear ratio for both the arm and the claw, significantly reducing strain on the motors, pulleys, and individual gear interactions. Throughout the competition, it also became clear that precision in final object placement mattered far more than we initially anticipated, and this ultimately contributed to our downfall in the grand final. A more robust claw would likely resolve this issue and reduce the time required to score the tennis balls in particular.

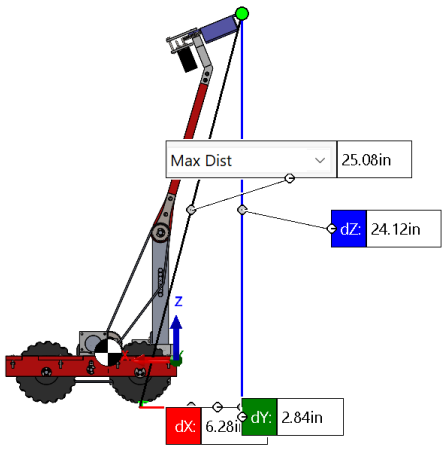
Despite these challenges, we remain proud of our weight optimization. We were the lightest robot in the field, yet we incorporated more aluminum than most teams. We minimized weight at every opportunity and used complex milling and turning operations to remove excess mass from aluminum components. Our mobility was another strong point, our robot moved quickly and climbed ramps and turf with ease.

Overall, this project pushed us to balance creativity, engineering rigor, and real-world constraints. Our robot performed well in speed, mobility, and weight optimization, and the challenges we encountered with tensioning, precision placement, and long-term mechanical reliability taught us the value of designing for serviceability and robustness rather than performance alone. These lessons will shape our future engineering decisions in meaningful ways. Even though we faced setbacks in the final round, we are proud of the system we built and the principles we applied. We leave the project with a stronger understanding of how to design, test, and refine complex mechanical systems under competitive pressure.

4 Final Design Specifications

Table 2: Final Design Specifications

Parameter	**Value**
Total mass	5.67 lb)
Overall volume	114.00 in^3
Exterior surface area	1154.32 in^2
Collapsed footprint (L $\times$ W)	10.5 in $\times$ 10.5 in
Collapsed height	10.98 in
Maximum reach	24.12 in
Undercarriage ground clearance h	2.6 in
Wheelbase length L	6.0 in
Maximum breakover angle $\theta = 2\tan^{-1}\left(\frac{2h}{L}\right)$	$\theta \approx 81.8^\circ$
Turning radius	0 in (tank-track pivot in place)
Drivetrain gear ratio	*4:1*
Arm gear ratio	*5.33:1*



(a) Height Check in SolidWorks



(b) Photo of the Robot in ”Box of Justice”

Figure 12: Measuring Final Design Specifications

5 Appendix

5.1 Bill of Materials

Item	Part Number	Material	Manufacturing	Qty
1	BasePlate1	Acrylic	Laser Cut	1
2	Mast	3D-Printed PLA	Mill	1
3	Unpowered Arm	Acrylic	Laser Cut	1
4	Right Claw Bracket	3D-Printed PLA	0.02 pounds 10% infill	1
5	Left Claw Bracket	3D-Printed PLA	0.02 pounds 10% infill	1
6	Motor Mounts-Standard	3D-Printed PLA	0.05 pounds 10% infill	1
8	Side of baseplate	Acrylic	Laser Cut	2
9	Wheels	Acrylic & Dragon Skin Silicon	Cast	4
10	16XL037-FS16-.25H1.5L0.2500N	3D-Printed PLA	0.0102 pounds 22% infill	4
11	12XL037-FS12-.25H1.5L0.2500N	3D-Printed PLA	0.0102 pounds 22% infill	2
12	Belt1-1 Bot Final	Rubber	Parts Bin	1
13	Tensioner shaft	6061 Aluminum	Lathe	1
14	Belt2-1 Bot Final	Rubber	Parts Bin	1
15	Left Claw	3D-Printed PLA	0.03 pounds 10% infill	1
16	Claw Baseplate	Acrylic	0.03 pounds 10% infill	2
17	Right Claw	3D-Printed PLA	0.03 pounds 10% infill	1
18	Inch-Spur gear 32DP 48T 14.5PA .2FWS48O.5H.5L0.25N	3D-Printed PLA	0.03 pounds 10% infill	5
19	CR-PHMS 0.112-48x0.75x0.75-N	6061 Aluminum	Parts Bin	10
20	MSHXNUT 0.112-48-S-N	6061 Aluminum	Parts Bin	10
21	Claw Drive Shaft	6061 Aluminum	Lathe	2
22	Servo	3D-Printed PLA	Parts Bin	1
23	CR-PHMS 0.099-56x0.625x0.625-N	6061 Aluminum	Parts Bin	5
24	MSHXNUT 0.125-44-S-N	6061 Aluminum	Parts Bin	6
25	Mast Shaft	6061 Aluminum	Lathe	1
26	B27.1 - NA3-25	6061 Aluminum	Parts Bin	11
27	16XL037-FS16-.375H.5L0.2500N	3D-Printed PLA	0.0226 pounds 22% infill	1
29	Aluminum-Shaft-Support	6061 Aluminum	Dril Press	3
30	Delrin-Shaft-Support	Delrin	0.043 pounds 10% infill	3
31	Hex-Shaft	6061 Aluminum	Lathe	3
33	Inch - Spur gear 32DP 16T 14.5PA .2FW — S16O.5H.4L0.25N	3D-Printed PLA	Parts Bin	3
34	Motor	–	Parts Bin	3
35	Planetary	6061 Aluminum	Parts Bin	3
36	12XL037-FS12-.375H.5L0.2500N	3D-Printed PLA	0.0102 pounds 22% infill	1

Item	Part Number	Material	Manufacturing	Qty
37	Belt3-1 Bot Final	Rubber	Parts Bin	1
38	Complex Drivetrain Motormount	3D-Printed PLA	0.09 pounds 10% infill	2
39	CR-PHMS 0.06-80x0.5x0.5-N	6061 Aluminum	Parts Bin	12
40	CR-PHMS 0.06-80x0.625x0.625-N	6061 Aluminum	Parts Bin	6
41	Regular FW 0.06	6061 Aluminum	Parts Bin	14
42	MSHXNUT 0.060-80-S-N	6061 Aluminum	Parts Bin	20
43	B27.7M - 3CM1-8	6061 Aluminum	Parts Bin	4
44	Motor Drive Shaft	6061 Aluminum	Lathe	2
45	B27.1 - NA3-S31	6061 Aluminum	Parts Bin	14
46	CR-PHMS 0.099-56x0.75x0.75-N	6061 Aluminum	Parts Bin	8
47	Narrow FW 0.099	6061 Aluminum	Parts Bin	11
48	MSHXNUT 0.099-56-S-N	6061 Aluminum	Parts Bin	7
49	Arm Support Bracket	3D-Printed PLA	0.03 pounds 10% infill	1
50	MSHXNUT 0.099-48-S-N	6061 Aluminum	Parts Bin	2
51	Mast Bracket	6061 Aluminum	Dril Press	1
52	Left Mast Bracket	6061 Aluminum	Drill Press	1
53	CR-PHMS 0.138-32x1.375x1-N	6061 Aluminum	Parts Bin	2
54	MSHXNUT 0.138-32-S-N	6061 Aluminum	Parts Bin	2
56	CR-FIMS 0.125-44x1.125x0.5-N	6061 Aluminum	Parts Bin	2
57	Narrow FW 0.06	6061 Aluminum	Parts Bin	2
58	CR-PHMS 0.06-80x0.375x0.375-N	6061 Aluminum	Parts Bin	2
59	Aluminum-Shaft-Support_front	6061 Aluminum	Drill Press	2
60	CR-PHMS 0.06-80x0.5625x0.5625-N	6061 Aluminum	Parts Bin	10
61	ShorterDelrinSupport	3D-Printed PLA	0.038 pounds 10% infill	1
62	Front Wheel Shaft	6061 Aluminum	Lathe	3
63	Bearing	Plastic	Parts Bin	11
64	Mast Tension Rod	6061 Aluminum	Lathe	7
65	Powered Arm	Acrylic	Laser Cut	1
66	Batteries	–	Parts Bin	1
67	RC Reciever	3D-Printed PLA	Parts Bin	1
68	Motor Controler	3D-Printed PLA	Parts Bin	3
69	CR-PHMS0.112-40x0.375x0.375-N	6061 Aluminum	Parts Bin	2
70	BumperPlate	Acrylic	Laser Cut	1
71	CR-PHMS 0.112-40x0.5x0.5-N	6061 Aluminum	Parts Bin	2
72	16XL037-FS16-.375H.5L0.2500N	3D-Printed PLA	0.0126 pounds 22% infill	1

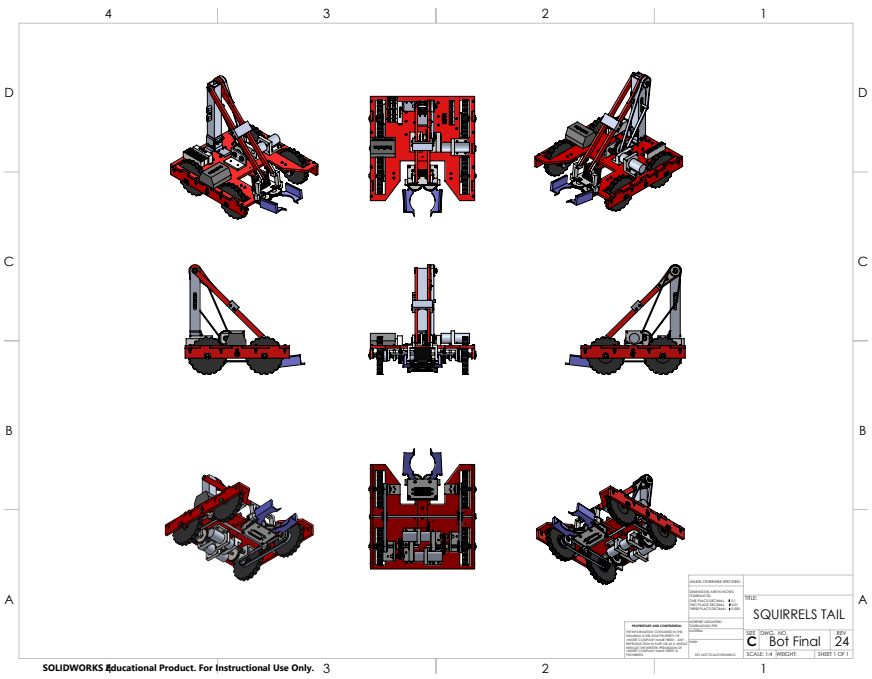


Figure 13: Overall robot final drawing.

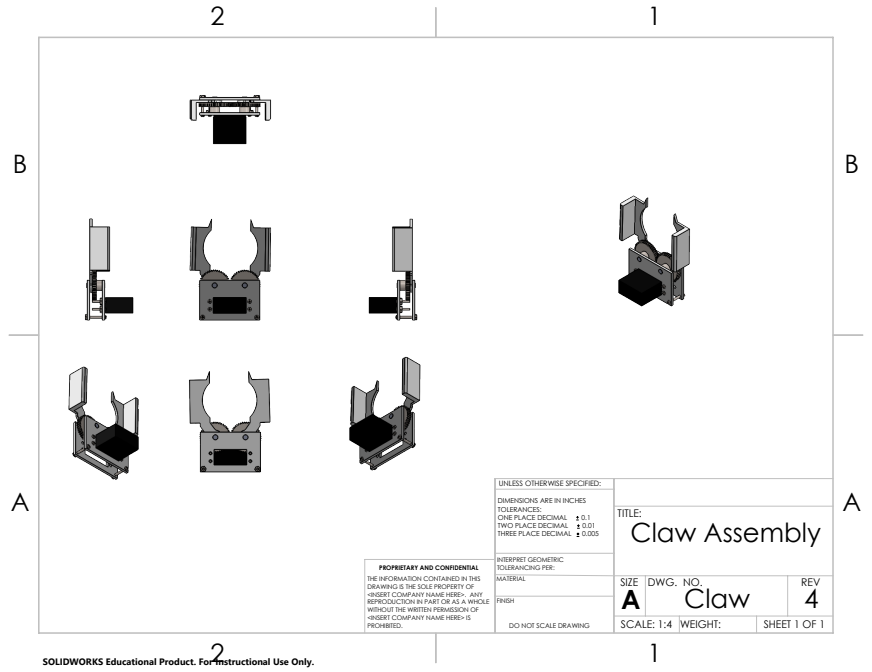


Figure 14: Claw assembly drawing.

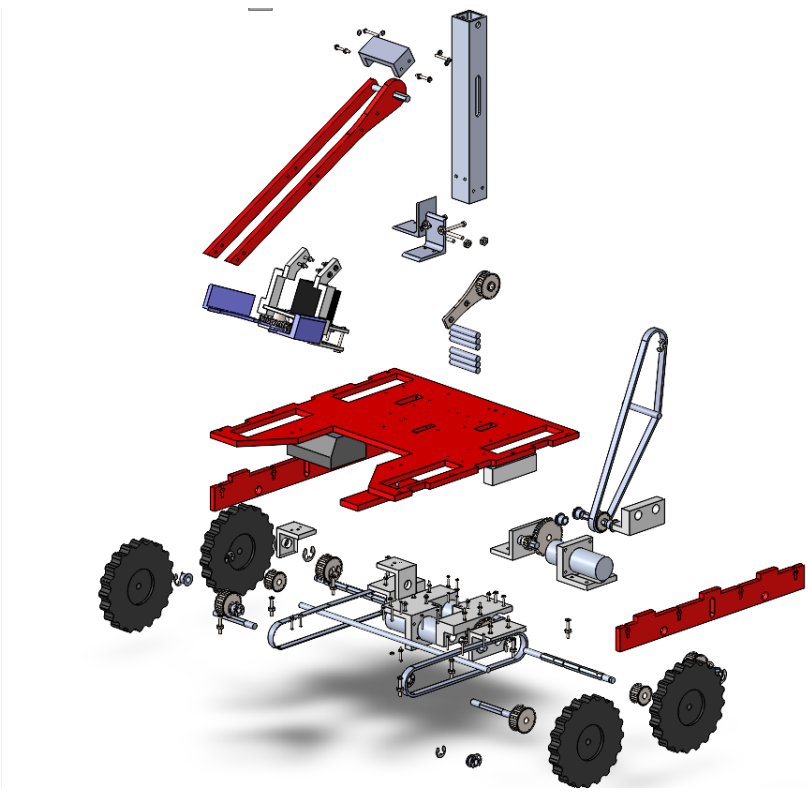


Figure 15: Exploded view of the assembly.

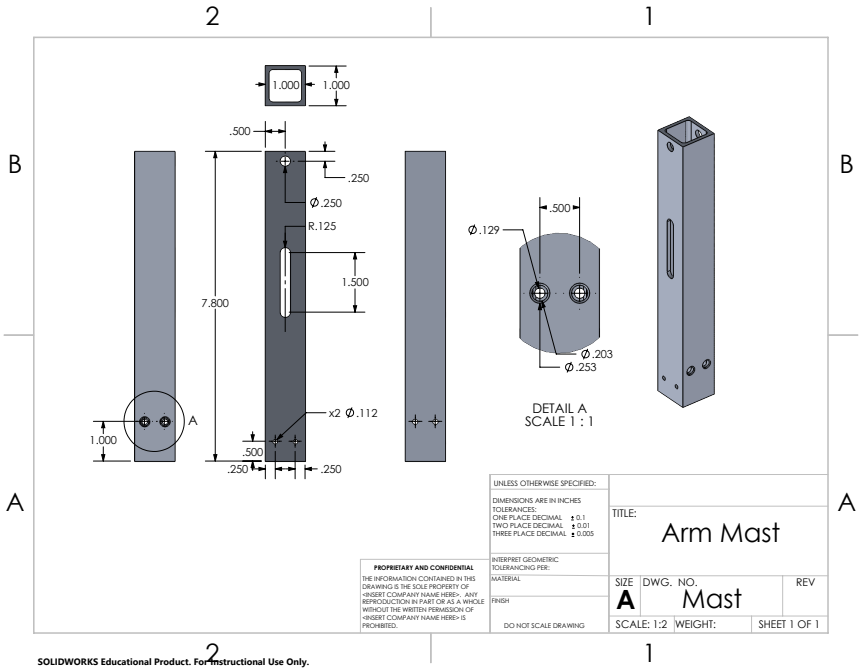


Figure 16: Mast drawing.

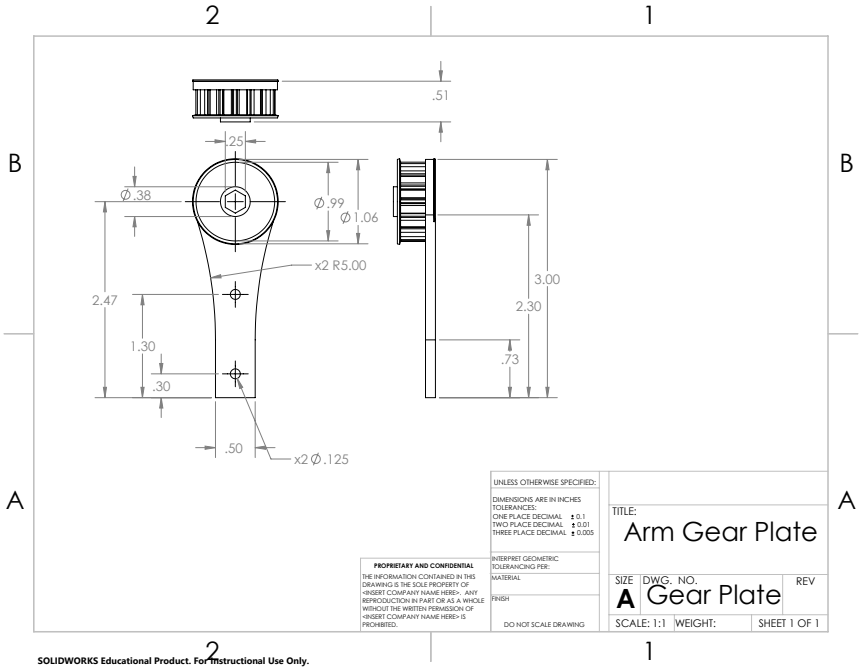


Figure 17: Timing pulley drawing.

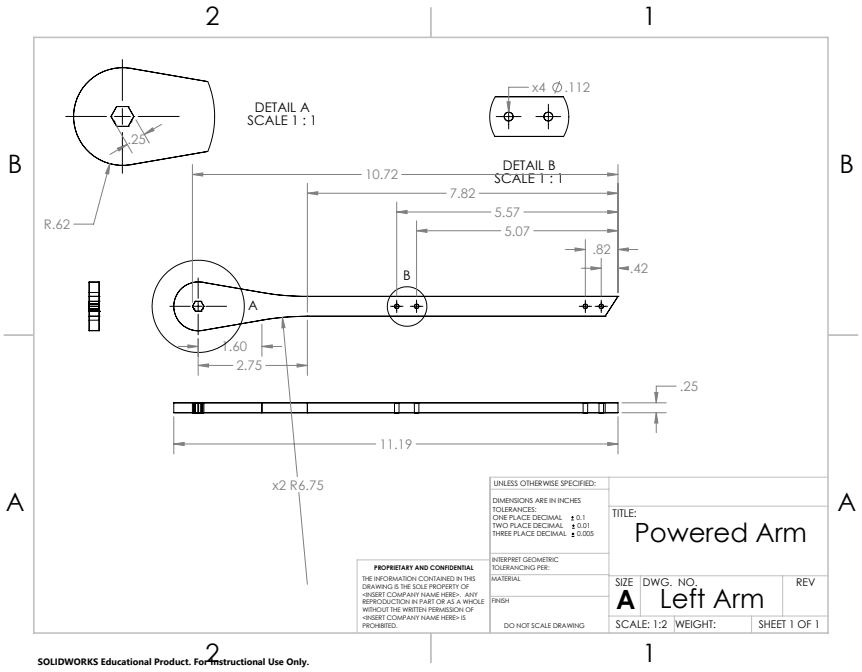


Figure 18: Powered arm drawing.

6 Contributions

Xavier contributed to the drivetrain calculations write-up and free body diagrams, the bill of materials, the drawing files, assembly files, the final design specifications, formatting and the overall reflection.

Caleb discussed success criteria, alternative solutions, final solutions, and measured the final design specifications. He also contributed to the overall reflection and figure/general formatting.

Diego contributed with the abstract, constraints, arm analysis calculations and its free body diagram, claw analysis, the motor characterization graphs, the beam analysis, formatting, and proofreading.

Rhianne contributed to creating first design for dumptruck and Pugh Matrix, calculating breaker angle design and calculations, and initial layout of the report and formatting.

William contributed to the idealized moment calculations and free body diagrams, drivetrain (ramp) calculations, alternative solutions (Pugh matrix), layout of the report, and formatting.